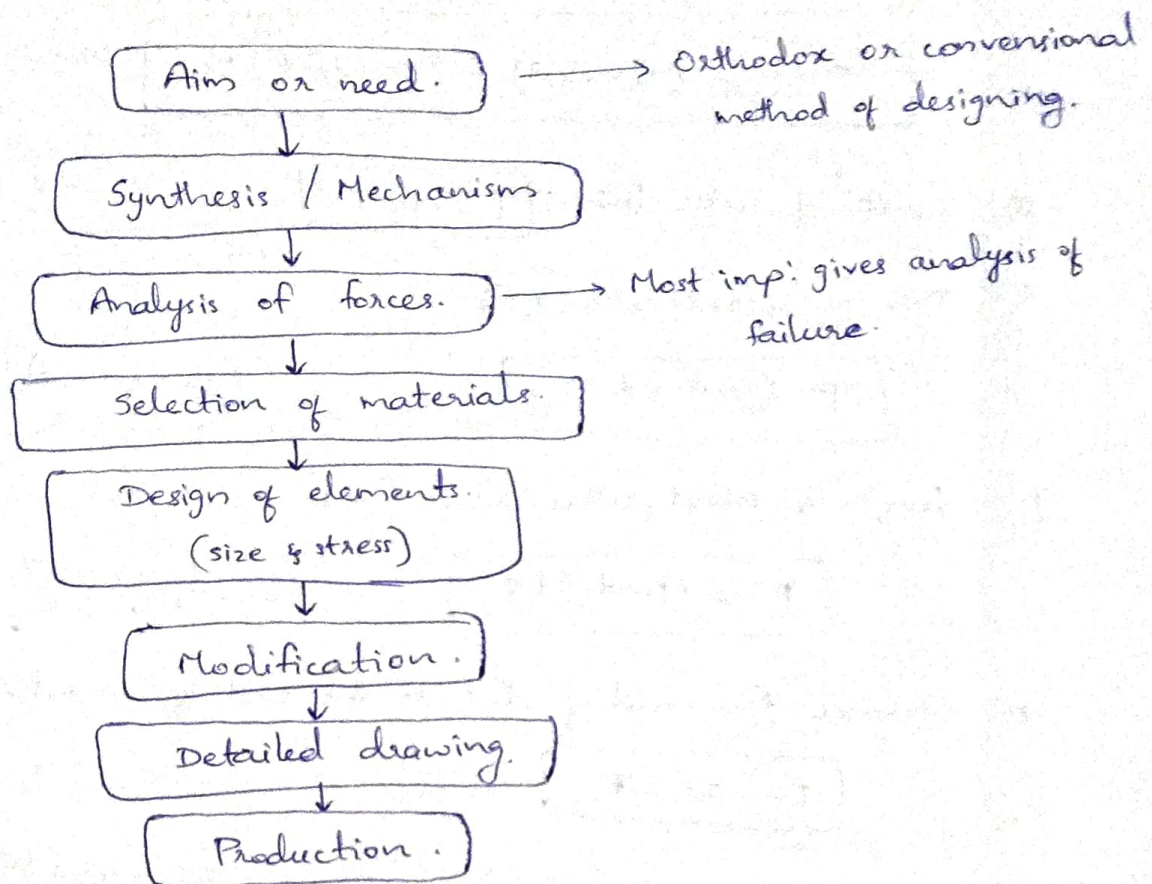


General Design Procedure

Now a days concurrent designing method is used.

Cotter jointsDesign procedure of cotter joint.

1. Diameter of rod $\Rightarrow P = \frac{\pi}{4} d^2 \sigma_t$ [Pg. 65, 4.16 (a)]
2. Dia of spigot & thickness of cotter $\Rightarrow P = \left[\frac{\pi}{4} d_1^2 - d_1 t \right] \sigma_t$ (4.16 (c))
 $t = 0.25 d_1$ (table P-66)

Find d_1 & t .

3. Crushing stress (σ_c) $\Rightarrow P = d_1 t \sigma_c$ (4.16 (b))

If the calculated $\sigma_c >^{\text{max.}}$ permissible σ_c ; design is not safe.

Then, If ~~calculated~~ we need to redesign d_1 & t by putting the safe some equation.

$$P = d_1 t \sigma_c$$

where, $\sigma_c \Rightarrow$ max permissible value.

[67 - 4.17 (a)]

4. Outer dia of socket D_1

$$P = \left[\frac{\pi}{4} (D_1^2 - d_1^2) - (D_1 - d_1) t \right] \sigma_t$$

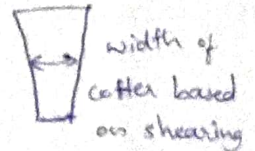
[67 - 4.18 (a)]

5. Width of collar (b) $\Rightarrow P = 2bt\tau$

6. Dia of socket collar (D)

$$P = (D - d_1) t \sigma_c$$

[67 - 4.17 (d)]



7. Length of socket collar (L)

$$P = 2(D - d_1) L \tau$$

[67 - 4.17 (c)]

τ' - not needed.

8. Distance from end of slot to the end of spigot, l . [65 - 4.16 (f)]

$$P = 2d_1 l \tau$$

9. Dia of spigot collar (d_2)

[65 - 4.16 (d)]

subjected more to crushing.

$$P = \frac{\pi}{4} (d_2^2 - d_1^2) \sigma_c$$

10. Thickness of collar, t ,

[65 - 4.16 (e)]

$$P = \pi d_1 t \tau$$

11. Bending moment

[67 - 4.18 (b)]

$$P = \frac{4}{3} \frac{tb^2 \sigma_b}{D}$$

If calculated $\sigma_b >$ Permissible σ_b , design not safe.

note: Find all other values from Pg-66.

- Design a cotter joint to support a load varying from 30kN in compression to 30kN in tension. The material used is carbon steel for which the following allowable stresses may be used.

$$\sigma_t = 50 \text{ MPa}, \quad \sigma_{\text{compressive}} = 50 \text{ MPa}.$$

$$\tau = 35 \text{ MPa (shear)}.$$

$$\sigma_c \Rightarrow \text{crushing} = 90 \text{ MPa}.$$

$$P = 30 \text{ kN}.$$

Ans. Dia of rod, $d \Rightarrow P = \frac{\pi}{4} d^2 \sigma_t \Rightarrow d = \sqrt{\frac{P \times 4}{\pi \sigma_t}}$

$$d = \sqrt{\frac{30 \times 10^3 \times 4}{\pi \times 50}} = \underline{28 \text{ mm}}$$

$\Rightarrow$ Dia of spigot (d_1) & t , $P = \left[\frac{\pi}{4} d_1^2 - d_1 t \right] \sigma_t$. — (1).

$$30 \times 10^3 = \left[\frac{\pi}{4} d_1^2 - d_1 t \right] 50.$$

$$\frac{\pi}{4} d_1^2 - d_1 t - 600 = 0 \text{ — (1)}.$$

$$\Rightarrow d_1 = \underline{34 \text{ mm}}.$$

$$t = 0.25 d_1 \Rightarrow 0.25 \times 34 = \underline{9 \text{ mm}}.$$

$\Rightarrow$ Crushing stress, $\sigma_c \Rightarrow P = d_1 t \sigma_c \quad ; \quad \sigma_c = 98 \text{ MPa}.$

$$\sigma_c = \frac{P}{d_1 t} = \frac{30 \times 10^3}{34 \times 9} = \underline{98 \text{ MPa}}$$

Permissible $\sigma_c = 90 \text{ MPa}$. $\therefore$ we have to redesign

$\therefore$ calculated $\sigma_c >$ permissible σ_c .

$$\therefore 30 \times 10^3 = d_1 t \times 90$$

$$\therefore d_1 = \underline{37 \text{ mm}}.$$

$$t = 0.25 d_1 = 0.25 \times 37 = \underline{10 \text{ mm}}$$

Now use new d_1 & t .

VII. $l = 13 \text{ mm}$

VIII. $l_1 = ?$

$$P = 2 d_1 l_1 \tau$$

$$30 \times 10^3 = 2 \times 37 \times l_1 \times 35$$

$$\therefore l_1 = \frac{30 \times 10^3}{2 \times 37 \times 35} = \underline{\underline{11.6}} \approx \underline{\underline{12 \text{ mm}}}$$

IX. $P = \frac{\pi}{4} (d_2^2 - d_1^2) \sigma_c$

$$30 \times 10^3 = \frac{\pi}{4} (d_2^2 - 37^2) 90$$

$$d_2^2 - 1369 = \frac{30 \times 10^3 \times 4}{\pi \times 90} = \underline{\underline{385.8}} \approx \underline{\underline{386.424.4}}$$

$$\therefore d_2 = \sqrt{\frac{386 + 1369}{424.4}} = 42.34 \approx \underline{\underline{43 \text{ mm}}}$$

X. $P = \pi d_1 t_1 \tau$

$$30 \times 10^3 = \pi \times 37 \times t_1 \times 35$$

$$\therefore t_1 = \frac{30 \times 10^3}{37 \pi \times 35} = 7.37 \approx \underline{\underline{8 \text{ mm}}}$$

XI. $P = \frac{4}{3} \frac{t b^2 \sigma_b}{D}$

$$\sigma_b = \frac{P \times 3 \times D}{4 t b^2} = \frac{30 \times 10^3 \times 3 \times 71}{4 \times 43^2 \times 10} = \underline{\underline{28.7}} \approx \underline{\underline{29 \text{ mm}}}$$

$$= 86.39 \approx \underline{\underline{87 \text{ N/mm}^2}}$$

$$\sigma_b \text{ permissible} = 90 \text{ N/mm}^2$$

XII.

XII. clearance = 2 mm



Design procedure of knuckle joint.

[Pg-67 4.19(a)]

1. Dia of rod, $d \Rightarrow P = \frac{\pi}{4} d^2 \sigma_t$

2. dia of pin, $d_1 \Rightarrow P = \frac{\pi}{2} d_1^2 \tau$

[67 - 4.20(a)]

$\rightarrow$ Compare with standard value.

3. Outer dia of eye, $D \Rightarrow P = F(D - d_1) \sigma_t$

[67 - 4.21(b)]

4. Check crushing stress, $\sigma_c \Rightarrow P = F d_1 \sigma_c$

[67 - 4.21(c)]

5. Width of square section, d_2 or s

[67 - 4.20(b)]

$$\Rightarrow P = d_2^2 \sigma_t$$

6. Check tensile stress, $\sigma_t \Rightarrow P = 2 A d_2 \sigma_t$

[67 - 4.19(b)]

7. Check tensile stress, $\sigma_t \Rightarrow$

[67 - 4.21(a)]

$$P = 2 (D - d_1) B \sigma_t$$

8. Check crushing stress, σ_c

[67 - 4.21(d)]

$$\Rightarrow P = 2 B d_1 \sigma_c$$

Note: Find all other values from pg-68 table.

1. Design a knuckle joint to transmit 150 kN. The design stresses may be taken as 75 MPa tension, 60 MPa shear, 150 MPa crushing.

Ans. Given, $P = 150 \text{ kN} \times 10^3 \text{ N}$.

$$\sigma_t = 75 \text{ MPa.}$$

$$\sigma_c = 150 \text{ MPa.}$$

$$\tau = 60 \text{ MPa.}$$

dia of rod, $d \Rightarrow P = \frac{\pi}{4} d^2 \sigma_t$

$$d = \sqrt{\frac{150 \times 10^3 \times 4}{75 \times \pi}} = 50.4 \approx \underline{\underline{51 \text{ mm}}}$$

dia of pin, $d_i \Rightarrow P = \frac{\pi}{2} d_i^2 \tau$

$$d_i = \sqrt{\frac{P \times 2}{\pi \times \tau}} = \sqrt{\frac{150 \times 10^3 \times 2}{\pi \times 60}} = 39.8 \approx \underline{\underline{40 \text{ mm}}}$$

$d_i = d$ from table, $\therefore d_i = d \Rightarrow 51 \text{ mm}$.

Outer dia of eye, $D \Rightarrow P = F (D - d_i) \sigma_t$.

$$F = 1.5d = 1.5 \times 51 = 76.5 = \underline{\underline{77 \text{ mm}}}$$

$$\therefore D = \frac{P}{F \times \sigma_t} + d_i$$

$$= \frac{150 \times 10^3}{77 \times 75} + 51 = 76.9 \approx \underline{\underline{77 \text{ mm}}}$$

$$D = 77 \text{ mm}.$$

Crushing stress, $\sigma_c \Rightarrow P = F d_i \sigma_c$

~~$$P = 150 \times 10^3 = F \times 51 \times 150$$~~

$$\therefore F = \frac{150 \times 10^3}{51 \times 150}$$

checking with std value from table,

$$D = 2d_i = 2 \times 51 = \underline{\underline{102 \text{ mm}}}$$

Crushing stress, $\sigma_c \Rightarrow P = F d_i \sigma_c$

$$\sigma_c = \frac{P}{F d_i} = \frac{150 \times 10^3}{77 \times 51} = \underline{\underline{392 \text{ N/mm}^2}}$$

width of square section, $s \Rightarrow P = [d_2^2 \sigma_t]$.

$$d_2 = s.$$

$$\frac{150 \times 10^3}{\sigma_t} = s^2.$$

$$s^2 = \frac{150 \times 10^3}{75} = 2000 \Rightarrow s = \sqrt{2000} = \underline{\underline{45 \text{ mm}}}$$

From table, $s = 1.2d = 1.2 \times 51 = 61.2 = \underline{\underline{62 \text{ mm}}}$

checking tensile stress, $\sigma_t \Rightarrow P = 2 A d_s \sigma_t$.

$$A = 0.6 d = 0.6 \times 51 = \underline{31 \text{ mm}}$$

$$\sigma_t = \frac{150 \times 10^3}{2 \times 31 \times 62} = \underline{40 \text{ N/mm}^2}$$

Again checking $\sigma_t \Rightarrow P = 2 (D - d_i) B \sigma_t$.

$$B = 0.75 d = 0.75 \times 51 = \underline{39 \text{ mm}}$$

$$\sigma_t = \frac{P}{2 (D - d_i) B} = \frac{150 \times 10^3}{2 (102 - 51) \times 39}$$

$$\sigma_t = \underline{38 \text{ N/mm}^2}$$

Crushing stress, $\sigma_c \Rightarrow P = 2 B d_i \sigma_c$.

$$\sigma_c = \frac{P}{2 B d_i} = \frac{150 \times 10^3}{2 \times 39 \times 51}$$

$$\sigma_c = \underline{38 \text{ N/mm}^2}$$

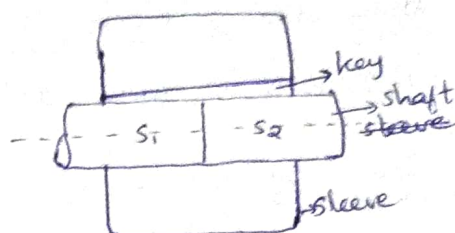
S	$1.2 d = 62 \text{ mm}$
F	$1.5 d = 77 \text{ mm}$
d_i	$d = 51 \text{ mm}$
C	$1.5 d = 77 \text{ mm}$
E	$d/2 = 26 \text{ mm}$
G_i	$3/16 d \text{ to } 1/4 d \Rightarrow 10 \text{ to } 13 \text{ mm}$
B	$0.75 d = 39 \text{ mm}$
H	$0.8 d = 41 \text{ mm}$
A	$0.6 d = 31 \text{ mm}$
l	$1.5 d \text{ to } 2 d \Rightarrow 77 - 102 \text{ mm}$
D	$2 d = 102 \text{ mm}$

L	$4 d = 204 \text{ mm}$
L_i	$4.8 d = 246 \text{ mm}$

4-1-24.

Designing of Coupling.Muff Coupling.

1. Design of shaft.
2. Design of muff / sleeve
3. Design key.



Based on torsional rigidity ;

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{l}$$

← based on power.

For designing solid shaft, $T = \frac{\pi}{16} \tau d^3$.Torque to be transmitted by shaft $\Rightarrow$

$$T = \frac{9.55 \times 10^6 P}{n} \quad \text{Nmm}$$

3.3a - Pg - 50.

 $n \Rightarrow \text{rpm}, P = \text{KW}.$

11-2-25

 $\rightarrow$ Step 1: design of shaft.

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{l}$$

 $J \Rightarrow$ Polar moment of inertia. $R \Rightarrow$ distance between central axis & outer surface of shaft.

$$R = \frac{D}{2}$$

$$T \text{ for circular cross section} = \frac{\pi}{32} d^4 \tau$$

$$P = \frac{2\pi NT}{60}$$

std. table

Pg. 57 3.5 (a).

$$T = \frac{9.55 \times 10^6 P}{N}$$

Step-2 : Design of muff or sleeve.

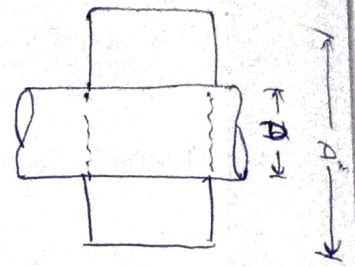
Considering sleeve as a hollow shaft.

Outer dia of sleeve, $D_m = 2D + 13 \text{ mm}$

Hub dia, $D_i = 1.5D$ [Pg-252].

Length of sleeve, $L = 2.5D + 50 \text{ mm}$ [Pg-253 : 13.2]

Ratio b/w inner & outer dia, $k = \frac{D_i}{D_o} = \frac{D}{D_m}$



Shear stress, τ , find it & check for safety. [Pg-50 : 3.1]

$$\tau = \frac{\pi T}{16 d_o^3} \left[\frac{1}{1-k^4} \right] \Rightarrow \tau = \frac{\pi}{16} \tau d_o^3 (1-k^4) \text{ for hollow shaft.}$$

$\tau_{calc} < \tau_{permissible}$; otherwise, design will fail.

Step-3 : Design of key [Rectangular key].

$$b = \frac{d}{4} \quad \text{eqn 4.2 (a) Pg-61}$$

$$h = \frac{d}{6} \quad \text{eqn 4.2 (b) "}$$

Taking dia of shaft, $D = 70$.

by std value, width, $b = 20$, height, $h = 12$.

Length of key based on crushing stress. $\therefore$

$$T = \frac{1}{4} \sigma_{b_1} h d l - \underbrace{\frac{1}{18} \sigma_{b_1} b l^2}_{\text{negligible } \therefore \text{small change}} \quad \text{Pg-62 : 4.4 (b).}$$

$$\therefore T = \frac{1}{4} \sigma_{b_1} h d l$$

No need to standardize.

l_τ $\Rightarrow$ length based on shear stress.

$$T = \frac{1}{2} \tau_1 b d l - \underbrace{\frac{1}{9} \tau_1 b l^2}_{\text{negligible.}}$$

Pg 62 - 4.5 (a).

$$T = \frac{1}{2} \tau \cdot b d l$$

Find l

compare l_o & l_t

select the largest l for muff coupler.

1. Design a muff coupler which is used to connect a steel shafts transmitting 40kW at 350rpm. The material for the shaft and key is carbon steel for which available shear and crushing stresses may be taken as 40 MPa and 80 MPa respectively. The material for muff is cast iron for which allowable shear stress is 15 MPa.

Ans. $D_{std} = 56 \text{ mm}$.

$P = 40 \text{ kW}$, $n = 350 \text{ rpm}$.

$$T = \frac{9.55 \times 10^6 P}{n} = \frac{9.55 \times 10^6 \times 40}{350} = \frac{1091428.57}{1} = 1091428.57$$

$$T = \frac{\pi}{16} \times 40 \times D^3$$

$$D^3 = \frac{T \times 16}{\pi \times 40} = \frac{1091428.57 \times 16}{\pi \times 40} = 51.7 \text{ mm}$$

$$\therefore \text{std } D = \underline{56 \text{ mm}}$$

$$\text{Outer dia of sleeve, } D_m = 2D + 13 = 2 \times 56 + 13 = \underline{125 \text{ mm}}$$

$$\text{Hub dia, } D_i = 1.5D = \underline{84 \text{ mm}}$$

$$\text{Length of sleeve, } L = 2.5D + 50 \text{ mm} = \underline{190 \text{ mm}}$$

$$k = \frac{D}{D_m} = \frac{56}{125} = \underline{0.448}$$

τ = Cast iron - muff, $\tau = 15 \text{ MPa}$.

$\tau =$

Flanged coupling.

12-2-25

→ Design of shaft. → step: 1

$$T = \frac{9.55 \times 10^6 \times P}{N}$$

[Pg-50 : 3.3 (a)]

$$T = \frac{\pi}{16} \tau D^3$$

Find D.

std D Pg-57 → Table 3.5 (a)

→ Step: 2.

Hub diameter

$$D_1 = 1.5 D + 25 \text{ mm or } 2D$$

Pg-252 : 13.1 (d).

$$T = \frac{\pi}{16} \tau D^3$$

Find D, std D → Page 251

→ Step: 3

$$\text{No. of bolt, } i = \frac{1}{40} D + 2$$

$$\text{Bolt diameter, } d = \frac{0.423 D}{\sqrt{i}} + 7.5 \text{ mm or } \frac{0.5 D}{\sqrt{i}}$$

(251-252)

$$\text{Dia of bolt circle, } D_2 = 3D.$$

Outside dia of flange, $D_2 = 4D$ [252 - 13.1 (f)]

Hub length, $L = 1.5D$ [252 - 13.1 (a)]

Flange thickness, $t = 0.35D + 9 \text{ mm}$ [252 - 13.1 (h)]

check τ bolt.

[Length of key = hub length.]
check for stresses.

Page 252 - 13.2 (c).

$$\tau = \frac{\tau \pi d^2 D_2}{8} \quad \text{And } \tau \text{ and check } \tau_{\text{calc}} < \tau_{\text{max}}$$

→. check bearing stress (σ) of bolts & flange [252 - 13.2 (d)]

$$\tau = \frac{\sigma_b i t d D_2}{2}$$

check shear stress for flange.

$$\tau = \frac{\tau \pi D_2^2 t}{2} \quad [252 - 13.2 (e)]$$

1. Design a cast iron protective type flange coupling to transmit 15 kW at 900 rpm from an electric motor to a compressor. The service factor may be assumed as 1.35 [ie., max torque is 35% greater than full load torque]. The following permissible stresses may be used.

$$\tau_{\text{shaft / bolt}} / \tau_{\text{key}} = 40 \text{ MPa.}$$

Crushing stress factor for bolt & key, $\sigma = 80 \text{ MPa}$.
flange.

Shear stress for cast iron, $\tau = 8 \text{ MPa}$

Draw a neat sketch.

Ans. $P = 15 \text{ kW}$, $N = 900 \text{ rpm}$. service factor = 1.35.

$T_{\text{max}} = 35\% > \text{full load torque.}$

$$T = \frac{9.55 \times 10^6 \times P}{N} = \frac{9.55 \times 10^6 \times 15}{900} = \underline{\underline{159.16 \text{ kN}}}$$

$$T_{\text{max}} = T \times 1.35 = 159.16 \times 1.35 = \underline{\underline{214.875 \text{ kN}}}$$

$$D^2 = \sqrt{\frac{T_{\text{max}} \times 16}{\pi \times \tau}} = \sqrt{\frac{214.875 \times 16}{\pi \times 40}} = \underline{\underline{30.19 \text{ mm}}}$$

$$D_{std} = \underline{32 \text{ mm}}$$

$$D_1 = 2D = 2 \times 32 = \underline{64 \text{ mm}}$$

$$i = \frac{1}{40} D + 2 = \frac{1}{40} \times 32 + 2 = \underline{2.8 \text{ mm}} \approx \underline{3}$$

$$\text{Bolt dia, } d = \frac{0.5 D}{\sqrt{i}} = \frac{0.5 \times 32}{\sqrt{3}} = \underline{9.23 \text{ mm}} \approx \underline{10 \text{ mm}}$$

$$\text{Bolt circle dia, } D_2 = 3D = 3 \times 32 = \underline{96 \text{ mm}}$$

$$\text{Outside flange dia, } D_3 = 4D = 4 \times 32 = \underline{128 \text{ mm}}$$

$$\text{Hub length, } L = 1.5 D = \underline{48 \text{ mm}}$$

$$\text{Flange thickness, } t = 0.35 D + 9 = (0.35 \times 32) + 9$$

$$= \underline{20.2} \approx \underline{21 \text{ mm}}$$

$$T = \frac{\tau \pi i d^2 D_2}{8} \Rightarrow \tau = \frac{8T}{\pi i d^2 D_2}$$

$$= \frac{18.99 \text{ Nmm}}{\pi \times 3 \times 21^2 \times 96} = \underline{18.99 \text{ MPa}} \Rightarrow \text{safe}$$

$$T = \frac{\sigma_b i t d D_2}{2} = \frac{\sigma_b \times 3 \times 21 \times 10 \times 96}{2}$$

$$\sigma_b = \frac{T_{max} \times 2}{3 \times 21 \times 10 \times 96}$$

$$= \underline{7.1 \text{ MPa}} \Rightarrow \text{safe}$$

$$T = \frac{\tau \pi D_1^2 t}{2} \Rightarrow \tau = \frac{T_{max} \times 2}{\pi D_1^2 t}$$

$$= \frac{1.6 \text{ MPa}}{\pi \times 64^2 \times 21} = \underline{1.6 \text{ MPa}} \Rightarrow \text{safe}$$

$$\text{Width of key, } b = \frac{D}{4} = \frac{32}{4} = 8 \text{ mm} \approx \underline{10 \text{ mm}}$$

$$\text{Height of key, } h = \frac{D}{6} = \frac{32}{6} = 5.3 \text{ mm} \approx \underline{8 \text{ mm}}$$

$$\text{Length of key} = \text{hub length} = 48 \text{ mm}$$

